$$F = ma$$

$$E = mc^2$$

# physics by log tables

Leaving Certificate Physics, read through the Formulae and Tables booklet

$$\rho = m/V$$

$$p = F/A$$

Definitions · units · experiments · derivations
Annotated for the new specification (first exam 2027)

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Click on the section you need

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Definitions of units

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Modern physics

Derivations

**Base units**

The international System of Units (Systeme International d'Unites) is founded on seven base quantities, which are assumed to be mutually independent. These base units are:

| Base quantity | SI base unit | Symbol for unit |
|---|---|---|
| length (l) | metre | m |
| mass (m) | kilogram | kg |
| time (t) | second | s |
| electric current (I) | ampere | A |
| temperature (T) | kelvin | K |
| amount of substance (n) | mole | mol |
| luminous intensity ($I_v$) | candela | cd |

**Derive units**

A derived unit is a unit which can be expressed in terms of base units and is given a unique name,

e.g. newton (N) = kg m $s^{-2}$.

**Prefixes**

Prefixes are used to form decimal multiples and submultiples of SI units. The common prefixes are:

| **Prefix** | Symbol | Factor | **Prefix** | Symbol | Factor |
|---|---|---|---|---|---|
| yotta | Y | $10^{24}$ | yocto | y | $10^{-24}$ |
| zetta | Z | $10^{21}$ | zepto | z | $10^{-21}$ |
| exa | E | $10^{18}$ | atto | a | $10^{-18}$ |
| peta | P | $10^{15}$ | femto | f | $10^{-15}$ |
| tera | T | $10^{12}$ | pico | p | $10^{-12}$ |
| giga | G | $10^{9}$ | nano | n | $10^{-9}$ |
| mega | M | $10^{6}$ | micro | μ | $10^{-6}$ |
| kilo | k | $10^{3}$ | milli | m | $10^{-3}$ |
| hecto | h | $10^{2}$ | centi | c | $10^{-2}$ |
| deka | d | $10^{1}$ | deci | d | $10^{-1}$ |

The symbol for a prefix is combined with the symbol for the base unit to form a new unit symbol, e.g. kilometre (km), microsecond (μs).

# Fundamental physical constants

| Constant | Value | Symbol | Comment |
|---|---|---|---|
| alpha particle mass | $6.644\,6565 \times 10^{-27}$ kg | $m_\alpha$ | **Used to determine number of particles in a mass of substance (i.e. 16g of O contains $N_A$ particles)** |
| Avogadro constant | $6.022\,1415 \times 10^{23}$ $mol^{-1}$ | $N_A$ | |
| Boltzmann constant | $1.380\,1415 \times 10^{-23}$ J $K^{-1}$ | k | **Unit of energy associated with atomic calculations** |
| electron mass | $9.109\,3826 \times 10^{-31}$ kg | $m_e$ | |
| electron volt | $1.602\,176\,53 \times 10^{-19}$ J | eV | |
| electronic charge | $1.602\,176\,53 \times 10^{-19}$ C | e | |
| Faraday constant | 96 485.3383 C $mol^{-1}$ | F | |
| gravitational constant | $6.6742 \times 10^{-11}$ $m^3$ $kg^{-1}$ $s^{-2}$ | G | |
| neutron mass | $1.674\,927\,28 \times 10^{-27}$ kg | $m_n$ | |

| Constant | Value | Symbol | Comment |
|---|---|---|---|
| permeability of free space | $4\pi$ x 10-7 H $m^{-1}$ | $\mu_0$ | **No speed should ever exceed $c_0$** |
| permittivity of free space | 8.854 187 817 x $10^{-12}$ F $m^{-1}$ | $\varepsilon_0$ | **Definition: one twelfth the mass of a carbon twelve atom (see Avogadro's number)** |
| Planck constant | 6.626 0693 x $10^{-34}$ J s | h | |
| proton mass | 1.672 621 71 x $10^{-27}$ kg | $m_p$ | |
| proton-electron mass ratio | 1836.182 672 16 | $m_p/m_e$ | |
| speed of light *in vacuo* | 2.997 924 58 x $10^{8}$ m $s^{-1}$ | $c_0$ | |
| unified atomic mass unit | 1.660 5402 x $10^{-27}$ kg | u | |
| universal gas constant | 8.314 472 J $K^{-1}$ $mol^{-1}$ | R | |

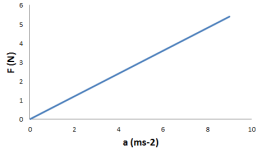
| Class name | Half-life | Mass | Symbol | Comment | | |
|---|---|---|---|---|---|---|
| **Leptons** | stable | 1 | e | **Defn:** Leptons do not feel strong force. | | |
| electron | stable | 4.305 x 10-6 | ν | Leptons are fundamental particles | | |
| neutrino | $1.52 \times 10^{-6}$ s | 206.9 | $\mu^+ \mu^-$ | | | |
| muon | $2.05 \times 10^{-13}$ s | 3478 | $\tau^+ \tau^-$ | | | |
| tau | stable | 0.3327 | $\nu_\mu$ | | | |
| muon neutrino | stable | 30.33 | $\nu_\tau$ | | | |
| tau neutrino | | | | | | |
| **Mesons** | $1.80 \times 10^{-8}$ s | 273 | $\pi^+ \pi^-$ | ┐ | ┐ | Has mass less than that of proton (2 quarks). |
| pi meson | $5.82 \times 10^{-17}$ s | 264 | $\pi^o$ | │ | │ | |
| K meson | $8.58 \times 10^{-9}$ s | 967 | $K^+ K^-$ | │ | ├ | |
| | ---- | 975 | $K^o$ | │ Hadrons feel | │ | |
| | | | | ┘ | ┘ | |
| **Baryons** | stable | 1836 | p | │ the strong | ┐ | Has mass greater than or equal to mass of proton (three quarks). |
| proton | $6.14 \times 10^{2}$ s | 1839 | n | ├ force. Consist | │ | Proton: $uud$ |
| neutron | $1.82 \times 10^{-10}$ s | 2183 | $\Lambda^o$ | │ of smaller | │ | Anti-proton: $\bar{u}\bar{u}\bar{d}$ |
| lambda | $5.56 \times 10^{-11}$ s | 2328 | $\Sigma^+$ | │ particles called | │ | Neutron: u$dd$ |
| sigma | $1.02 \times 10^{-10}$ s | 2343 | $\Sigma^-$ | │ quarks | ├ | Anti-neutron: $\bar{u}\bar{d}\bar{d}$ |
| chi | $5.13 \times 10^{-20}$ s | 2334 | $\Sigma^o$ | │ | │ | |
| omega | $1.14 \times 10^{-10}$ s | 2586 | $\Xi^-$ | │ | │ | |
| | $2.01 \times 10^{-10}$ s | 2573 | $\Xi^o$ | │ | │ | |
| | ~ $10^{-10}$ s | 3272 | $\Omega^-$ | ┘ | ┘ | |

| Quark | Charge | symbol | Comment |
|---|---|---|---|
| up | $\frac{2}{3}$ | u | All baryons made up of quarks. |
| strange | $-\frac{1}{3}$ | s | Mesons consists of 2 quarks (quark and antiquark) |
| top | $\frac{2}{3}$ | t | Baryons consist of three quarks. |
| down | $-\frac{1}{3}$ | d | Sum of quark charges add up to charge of particle, i.e. proton (+1 charge): uud (+2/3 +2/3 -1/3 = +1) |
| charmed | $\frac{2}{3}$ | c | neutron (0 charge): udd (+2/3 -1/3 -1/3 = 0) |
| bottom | $-\frac{1}{3}$ | b | anti-particles made up of antiquarks, i.e. |
| | $\frac{2}{3}$ | | anti-proton: $\bar{u}\bar{u}\bar{d}$ (-2/3 -2/3 +1/3 = -1) |
| | $-\frac{1}{3}$ | | anti-neutron: $\bar{u}\bar{d}\bar{d}$ (-2/3 +1/3 +1/3 = 0) |

An alphabetical list of the symbols used in the following formulae and their meaning in the relevant context is given on page 65.

Mechanics

**+ New** Spec 1.1 adds vectors: scalars vs vectors, resolving into perpendicular components, the resultant of two vectors (verify vector addition with data).

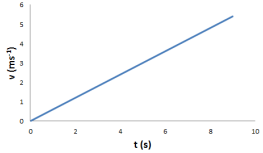
| | | |
|---|---|---|
| force and acceleration | $F = ma$ | **Defn:** Force is that which causes acceleration<br>**Expt:** Show $F \alpha a$<br>  <br><br>**Slope = mass (m)**<br>**Note: mass must remain constant (i.e mass on pan + mass of cart)**<br>**Deriv:** Show F=ma is special case of $F \alpha \frac{dp}{dt}$ (Newtons Second law)<br>✓ **On spec** Spec 1.2 lists Fnet = ma, and verifying Newton's 2nd law is a primary-data investigation. Deriving F = ma from momentum is not listed: check. |

**Linear motion with constant acceleration**

$$v = u + at$$

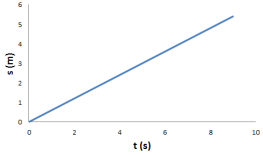
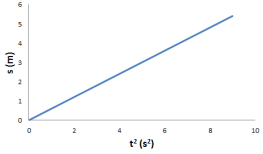
**Defn:** acceleration is rate of change of velocity

**Expt:** Measurement of acceleration ($v = at + u$)




**Slope = a**

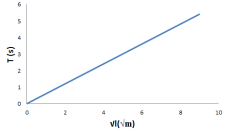
✓ **On spec** Spec 1.1: derive the kinematic equations; a = Δv/Δt and v = Δs/Δt are listed. Experiments become "investigate constant and varying linear motion".

| | | |
|---|---|---|
| | $s = ut + \frac{1}{2}at^2$ | **Deriv:** need to get rid of v. Sub $v = u + at$ into $s = \left(\frac{u+v}{2}\right)t$<br>**Expt:** Measurement of velocity ($s = ut$)<br>  <br>**Slope = v**<br>**Exp:** Measurement of accl. due to gravity ($s = \frac{1}{2}gt^2$)<br>  <br>**Slope = g/2** |
| | $v^2 = u^2 + 2as$ | **Deriv:** need to get rid of t. Sub $t = \frac{v-u}{a}$ into $s = \left(\frac{u+v}{2}\right)t$ |
| | $s = \left(\frac{u+v}{2}\right)t$ | **Defn:** average vel = total distance/total time |
| **Relative motion** | | |
| relative displacement | $\vec{s}_{BC} = \vec{s}_B - \vec{s}_C$ | |
| relative velocity | $\vec{v}_{BC} = \vec{v}_B - \vec{v}_C$ | |
| relative acceleration | $\vec{a}_{BC} = \vec{a}_B - \vec{a}_C$ | |

**Collisions**

| | | |
|---|---|---|
| momentum of a particle | $mv$ | **Defn:** momentum is product of mass and velocity |
| Newton's experimental law | $v_1 - v_2 = -e(u_1 - u_2)$ | |
| conservation of momentum | $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ | **Expt:** Verify principle of conservation of momentum (calculation)<br>**NB:** cannot write this eqn for priniciple of conservation of momentum<br>**Defn:** Momentum before a collision if equal to momentum after provided there is no external force<br>✓ **On spec** Spec 1.2. Now also two-dimensional collisions, using perpendicular and parallel components. |
| Impulse | $I = \int F\,dt = mv - mu$ | Newton's 2nd Law is here(ish). $F = \frac{mv-mu}{t}$<br>Deriv: Show $F = ma$ is a special case of Newton's 2n Law $\left(F\alpha\frac{\Delta p}{\Delta t}\right)$<br>× **Not on spec** Impulse is not listed in the specification. |
| **Motion in a circle** | | ? **Check** Only F = mv²/r is on the specification equations list. Radians, ω = θ/t, v = rω and centripetal acceleration are not listed, though outcome 1.6 says "constant angular velocity": check. |
| angle in radians | $\theta = \frac{s}{r}$ | Start point for derivation of $v = r\omega$ |
| angular velocity | $\omega = \frac{\theta}{t}$ | **Defn:** angular velocity is angle per unit time |
| linear and angular velocity | $v = r\omega$ | **Deriv:** $s = r\theta \rightarrow s/t = (r\theta)/t \rightarrow v = r\omega$ |
| centripetal acceleration | $a = r\omega^2 = \frac{v^2}{r}$ | |
| centripetal force | $F = mr\omega^2 = \frac{mv^2}{r}$ | **Defn:** centripetal force is force towards center that keeps a body moving in a circle. Let it equal the force that provides the centripetal force for derivation (e.g. see rotating bodies)<br>✓ **On spec** Spec 1.6, written F = mv²/r. |

**Rotating bodies**

| | | |
|---|---|---|
| angular momentum | $L = I\omega = rmv$ | |
| moment of a force | $M = Fd$ | **Defn:** moment is product of force and **perpendicular** distance<br>× **Not on spec** Moments are not on the specification. |
| torque of a couple | $T = Fd$ | **Defn:** torque of a pair of equal and opposite forces is **force of one times distance between them.**<br>× **Not on spec** Couples are not on the specification. |
| Newton's 2nd law of rotation | $T = \frac{dL}{dt}$ | |
| rotational kinetic energy | $E = \frac{1}{2}I\omega^2$ | |
| **Simple harmonic motion** | $a = -\omega^2 s$<br>$T = \frac{1}{f} = \frac{2\pi}{\omega}$<br>$s = A\sin(\omega t + \alpha)$<br>$v^2 = \omega^2\left(A^2 - s^2\right)$ | **Defn:** SHM when acc is proportional to displacement from equilibrium position<br>× **Not on spec** Simple harmonic motion is not on the specification. |
| simple pendulum | $T = 2\pi\sqrt{\frac{l}{g}}$ | **Expt:** Investigate relationship of period and length of simple pendulum<br>  <br>**Slope = $2\pi/\sqrt{g}$**<br>✓ **On spec** Spec 1.5, as $g = 4\pi^2 l/T^2$: one of the models for g (one by primary data, all four by secondary data). |
| compound pendulum | $T = 2\pi\sqrt{\frac{I}{mgh}}$ | |



Contents

| **Energy and work** | | **+ New** Spec 1.4 adds elastic potential energy, $E_p = \frac{1}{2}kx^2$, and work done in stretching or compressing. |
|---|---|---|
| Work | $W = Fs = \int F\,ds$ | **Defn:** work is the product of the force and displacement |
| Power | $P = \frac{W}{t} = Fv$ | **Defn:** power is work per unit time<br>✓ **On spec** P = W/t is on the specification list; P = Fv is not. |
| percentage efficiency | $\frac{P_o \times 100}{P_i}$ | ✓ **On spec** Efficiency is in the specification (2.1, and transformer efficiency in strand 3). |
| potential energy (gravitational) | $E_p = mgh$ | |
| kinetic energy | $E_k = \frac{1}{2}mv^2$ | |
| principle of conservation of energy (under conservative mechanical forces) | $\Delta E_p + \Delta E_k = 0$ | |
| mass-energy equivalence | $E = mc^2$ | |

**Gravitation**

**+ New** Spec 1.5 adds escape velocity, $v_e = \sqrt{(2GM/r)}$.

| | | |
|---|---|---|
| Newton's law of gravitation | $F = \frac{Gm_1m_2}{d^2}$ | **Defn:** Force of attraction between two point masses is proportional to the product of the masses and inversely proportional to the distance between them squared (inverse square law) |
| Weight | $W = mg = V\rho g$ | |
| acceleration due to gravity | $g = \frac{Gm}{d^2}$ | **Deriv:** $W = mg = \frac{GMm}{d^2}$ |
| gravitational field strength | $g = \frac{F}{m}$ | |
| period of a satellite | $T^2 = \frac{4\pi^2R^3}{GM}$ | Deriv: Circular motion $\rightarrow$ centripetal force = Force due to gravity $\frac{mv^2}{r} = G\frac{Mm}{r^2}$ then use, $T = 2\pi R/v$ ✓ **On spec** Spec 1.6: verify Kepler's 3rd law with secondary data; near-Earth and geostationary orbits. |

**Forces and materials**

| | | |
|---|---|---|
| Hooke's law | $F = -ks$ | **Defn**: Hooke's law – Restoring force is proportional to displacement from equilibrium position<br>**Defn:** Elastic constant is the ratio of applied force to extension (Force per unit length of extension)<br>**Deriv**: Show a system which obeys hooke's law moves with SHM<br>$F = -ks \rightarrow ma = -ks \rightarrow a = -\frac{k}{m}s$<br>$a\ \alpha - s$ implies SHM with $\omega^2 = k/m$<br>✓ **On spec** Spec 1.3, written F = −kx; verify Hooke's law with data. |
| stress | $\sigma = \frac{F}{A}$ | |
| strain | $\varepsilon = \frac{\Delta l}{l}$ | |
| Young's modulus | $E = \frac{\sigma}{\varepsilon}$ | |
| density | $\rho = \frac{m}{V}$ | **Defn**: density is mass per unit volume<br>✓ **On spec** Spec 1.2: "investigate density". |
| coefficient of friction | $\mu = \frac{F}{R}$ | |
| pressure | $p = \frac{F}{A}$ | **Defn**: pressure is force per unit area |
| pressure in a fluid | $p = \rho gh$ | ✓ **On spec** Spec 1.2, written P = hpg. |
| thrust on an immersed plane surface | $T = Ap_c$ | |
| Boyle's law | $p \propto \frac{1}{V}$ | **Defn**: Bolye's Law: pressure inversely proportional to volume **for fixed mass of gas at fixed temperature**<br>× **Not on spec** Boyle's law is not in the specification. |

**Heat and temperature**

**+ New** Spec 2.1 adds thermometric properties of materials (analyse thermometer suitability with data), the heat pump, efficiency and insulation/sustainability.

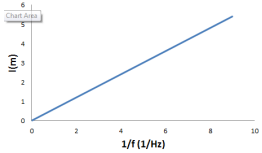
| | | |
|---|---|---|
| Celsius temperature | $\theta/^\circ\text{C} = T/K - 273.15$ | **Defn**: Celsius Scale / Kelvin Scale |
| energy needed to change temperature | $\Delta E = mc\Delta\theta$<br>$\Delta E = C\Delta\theta$ | **Defn**: $c = \frac{\Delta E}{m\Delta\theta}$ ,specific heat capacity is the energy per unit mass per unit temperature change.<br>**Defn**: $C = \frac{\Delta E}{\Delta\theta}$ ,heat capacity is the energy per unit mass per unit temperature change. |
| energy needed to change state | $\Delta E = ml$ $\Delta E = L$ | **Defn**: $l = \frac{\Delta E}{m}$ ,latent is the energy per unit mass require to change state of a substance (**e.g. fusion and vaporisation**).<br>**Defn**: $L = \Delta E$, energy to change state of a body |
| thermal conductivity | $\frac{\Delta E}{\Delta t} = kA\frac{\Delta\theta}{\Delta l}$ | |
| thermal resistivity | $r = \frac{1}{k}$ | |
| R-value (thermal resistance) | $R = \frac{l}{k} = lr$ | |
| U-value (thermal transmittance) | $\frac{\Delta E}{\Delta t} = AU\Delta\theta$ | $U = \frac{\Delta E}{\Delta t A \Delta\theta}$<br>U-value of a material is the Energy that passes through it per unit time per unit Area per unit temperature difference of the regions on each side<br>✓ **On spec** Spec 2.1: "U-value as the rate of transfer of heat through a material". |
| U-value of a composite barrier | $U = \frac{1}{\sum R}$ $\frac{1}{U} = \frac{1}{U_1} + \frac{1}{U_2} + \ldots$ | |



Contents

Light and sound

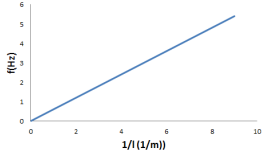
**+ New** Spec 2.4–2.5 add the EM spectrum, dispersion, solar irradiance and ultrasound. Intensity I = P/A is now on the equations list.

| | | |
|---|---|---|
| velocity of a wave | $c = f\lambda$ | **Expt**: Measure the speed of sound in air<br>$\lambda = c\,1/f$, $\lambda = 4l \rightarrow l = (c/4)\ (1/f)$<br>  <br><br>**Slope = $c/4$**<br>**Note**: line doesn't really pass through (0,0) but cuts the l-axis at (-0.3d). This means that the real distance from node to antinode is $l + 0.3d$ and $\lambda = 4\,(\,l + 0.3d\,)$. $c = f\lambda$ becomes $\mathbf{c = f4(l + 0.3d)}$<br>**? Check** v = fλ and T = 1/f are on the spec (2.2). Measuring the speed of sound is not a listed outcome: check. |
| Doppler effect | $f' = \dfrac{fc}{c \pm u}$ | Prefer the notation $f' = \left(\frac{c}{c \pm u}\right) f$ or $\frac{f'}{f} = \frac{c}{c \pm u}$.<br>**Defn**: apparent change in frequency due to relative motion between source and observer.<br>✓ **On spec** On the specification equations list (2.7). |

fundamental frequency of a stretched string

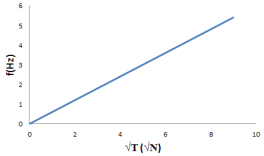
$$f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$$

**Expt 1**: Investigate variation of frequency with length of a stretched string




**slope** = $\frac{1}{2}\sqrt{T/\mu}$

Expt 2: Investigate variation of frequency with tension of a stretched string

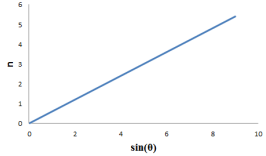



**Slope** = $1/(2l\sqrt{\mu})$

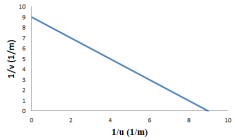
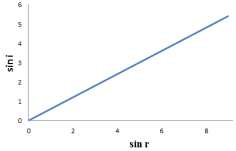
✓ **On spec** Spec 2.6: model standing waves on a stretched string and verify for a particular string (frequency vs length). Frequency vs tension: check.

refractive index

$$n = \frac{c_1}{c_2}$$

| | | |
|---|---|---|
| diffraction grating | $n\lambda = d \sin\theta$ | **Expt**: Measure wavelength of monochromatic light<br>Calculations and get average or graph below<br>  <br><br>**Slope = $d/\lambda$**<br>$$\textbf{Note: } 400\frac{\textbf{lines}}{\textbf{mm}} \rightarrow d = \frac{1mm}{400} = \frac{1\times10^{-3}}{400}m$$<br>**? Check** Diffraction and interference are in outcome 2.3, but nλ = d sin θ is not on the equations list and the wavelength experiment is not a listed outcome: check. |
| energy of a photon | $E = hf$ | ✓ **On spec** Spec 4.3 uses $E_2 - E_1 = hf$ (emission spectra). |
| Einstein's photoelectric law | $hf = \phi + \frac{1}{2}mv^2;\ \phi = hf_0$ | **? Check** The photoelectric effect is in strand 4, and the equations list gives threshold frequency $f_0$. Whether $hf = \varphi + \frac{1}{2}mv^2$ is required: check. |

**Geometrical optics**

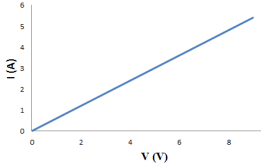
| | | |
|---|---|---|
| mirror and lens formula | $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ | **Expt**: Determine the focal length of a concave mirror or convex lens ( $1/v = -1/u + 1/f$)   ✓ **On spec** Spec 2.3, for lenses: converging lenses by primary data, diverging lenses by secondary data. × **Not on spec** Spherical mirrors (and the concave-mirror experiment) are not on the specification. |
| Magnification | $m = \frac{v}{u}$ | **Y-intercept =** $1/f$ ? **Check** Not on the specification equations list: check. |
| power of a lens | $P = \frac{1}{f}$ | **Note:** f must be in SI units. P is always given in ($m^{-1}$) × **Not on spec** Not on the specification. |
| two thin lenses in contact | $P = P_1 + P_2$ | × **Not on spec** Not on the specification. |
| refractive index | $n = \frac{\sin i}{\sin r} = \frac{1}{\sin C}$ | **Defn:** part 2 of laws of refraction **Expt:** Measurement of the refractive index of a block of glass   **Slope = n** ✓ **On spec** Spec 2.3, with critical angle and total internal reflection; verify refraction with data. |

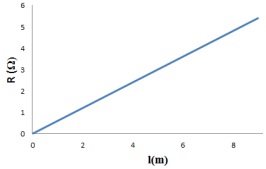


Contents

Electricity

**+ New** Draft strand 3 adds fuses and circuit breakers, p-n junctions, the n-p-n transistor, the DC motor, AC/DC generators and the National Grid.

| | | |
|---|---|---|
| Coulomb's law | $F = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{d^2}$ | **Defn:** Force of attraction or repulsion between two point charges is proportional to the product of the charges and inversely proportional to the distance between them squared (inverse square law) |
| electric field strength | $E = \frac{F}{q}$ | **Defn:** Electric field strength is the force per unit charge at that point |
| potential difference | $V = \frac{W}{q}$ | **Defn**: The potential difference between two points is the work done in bringing a unit charge from one point to the other. **(force per unit charge)** |
| resistance | $R = \frac{V}{I}$ | **Defn:** Resistance is the ratio of voltage to current<br>**Expt:** Investigate the variation of current with voltage for a metallic conductor **(ohmic conductor)**<br>  <br>**slope = $1/R$** |

| | | |
|---|---|---|
| resistivity | $\rho = \frac{RA}{l}$ | **Defn:** The resistivity of a material is the resistance that a cube of side 1m of that material will have.<br>**Expt**: Measure the resistivity of a wire<br>  <br>**Slope = ρ/A**<br>✓ **On spec** Spec list: R = ρL/A. Draft adds the effect of temperature on resistance (investigation). |
| resistors in series | $R = R_1 + R_2$ | **Deriv**: What's different: $V = V_1 + V_2$<br>What's the same: $I = I_1 = I_2$<br>Use ohm's law: $IR = I_1R_1 + I_2R_2$<br>Cancel to get: $R = R_1 + R_2$ |
| resistors in parallel | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ | **Deriv**: What's different: $I = I_1 + I_2$<br>What's the same: $V = V_1 = V_2$<br>Use ohm's law: $V/R = V_1/R_1 + V_2/R_2$<br>Cancel to get: $1/R = 1/R_1 + 1/R_2$ |
| Wheatstone bridge | $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ | **? Check** Not on the equations list or in the draft outcomes; likely removed. Check. |

Joule's law

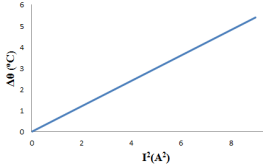
$P \alpha I^2$

**Defn**: $P \alpha I^2$ for constant $R$

**Expt**: Verification of Joule's Law

$$Power = \frac{Energy}{time} = \frac{mc\Delta\theta}{t}$$

Graph $\Delta\theta$ vs $I^2$ for constant $m,\ c,\ t$




High tension lines: Max V $\rightarrow$ Min I $\rightarrow$ Min P losses

**? Check** P = I²R and W = I²Rt are on the equations list. Verifying Joule's law is not a listed outcome in the draft: check.

| | | |
|---|---|---|
| power (instantaneous) | $P = VI$ | |
| force on a current-carrying conductor | $F = IlB; \quad l \perp B$ | **Note**: l must be $\perp$ B. If not $F = IlBsin\theta$<br>**? Check** Draft: investigate the force on a current-carrying wire, the motor effect and the DC motor. F = IlB is not legible on the equations list: check. |
| force on a charged particle | $F = qvB; \quad v \perp B$ | **Deriv**: $F = IlB,\ I = Q/t,\ Q = Nq$<br>$F = \frac{Nq\mathbf{l}B}{\mathbf{t}} = NqvB, \quad N = 1$<br>**? Check** Draft MP1: production and deflection of an electron beam. Check whether F = qvB is required. |
| induced emf | $E = -\frac{d\Phi}{dt}$ | **Defn**: Induced emf is prop to the rate of change of flux **(Farday's Law)**.<br>Minus sign because induced emf is always as to oppose the change that's causing it **(Lens' Law)**<br>Law is only for a single loop. If N loops then<br>$E = -N\frac{d\Phi}{dt}$ |
| alternating voltage and current | $V_{rms} = \frac{V_0}{\sqrt{2}} \qquad I_{rms} = \frac{I_0}{\sqrt{2}}$ | |
| capacitance | $C = \frac{q}{V}$ | **Defn**: Capacitance is the ratio of charge on a body to its potential<br>**? Check** Capacitors are not in the draft outcomes or on the equations list; likely removed. Check. |
| parallel-plate capacitor | $C = \frac{\varepsilon_0 A}{d}$ | Factors on which a capacitor depends. **A is the overlapping area** |

| | | |
|---|---|---|
| energy stored in capacitor | $W = \frac{1}{2}CV^2$ | Also by substitution: $W = \frac{1}{2}q^2/C$ |
| magnetic flux | $\Phi = BA$ | **Defn**: Magentic flux is the product of the magnetic flux density and $\perp$ area to the field lines. If not $\perp$ then $\Phi = BAsin\theta$<br>**? Check** Flux is used in the induction outcomes, and E = −dΦ/dt is on the equations list. Check whether Φ = BA is needed. |
| transformer | $\frac{V_i}{V_o} = \frac{N_p}{N_s}$ | ✓ **On spec** Draft: model transformers and solve problems on their efficiency. |

**Radioactivity**

**+ New** Draft strand 4 also has thermionic emission, X-ray production, emission spectra ($E_2 - E_1 = hf$), background radiation, Cockcroft and Walton, fission, fusion and particle–antiparticle interactions.

| | | |
|---|---|---|
| activity | $A = -\frac{dN}{dt}$ | **Defn**: number of radioactive emissions per unit time |
| law of radioactive decay | $A = -\lambda N$ | N is number of un-decayed nuclei. **If given a mass use avagadro's number to get N** |
| half-life | $T_{1/2} = \frac{\ln 2}{\lambda}$ | |
| mass-energy equivalence | $E = mc^2$ | |

# Definitions of units

**Definition of a unit:** If you are asked for the definition of a unit you hit the 1,1,1 definition.

**Examples**

**Definition of the volt:** The potential difference between two points is **1V** if the work done in bringing **1C** from one point to the other is **1J.**

**Definition of the Newton:** When a Force of **1N** is applied to a body of mass **1kg** then the resultant acceleration is **1ms$^{-2}$.**

**The Farad:** The capacitance of a body is **1F** if when a charge of **1C** is placed on it, the potential rises by **1V**.

Etc……..

**EXCEPT FOR THE DEFINITION OF THE AMPERE!!!!!**

The **ampere** is that constant current which, if maintained in two

- Infinite length wires
- Of negligible cross-section
- Placed one meter apart
- In a vacuum

Will exert a force on each other of $2 \times 10^{-7}$N per metre length

Note: The definition of the ampere has now changed to $I = Q/t$. **1 Ampere** is the current if a **1C** of charge passes in **1s**

# Equations not in the log tables

$I = P/A$ – Intensity (sound or light) ✓ **On spec** Now on the specification equations list (2.4).

$$v = \sqrt{\frac{GM}{d}}$$

– linear velocity of a orbiting plant (needs to be derived by letting $F = \frac{GMm}{d^2} = \frac{mv^2}{d}$)

For a mass on a spring executing SHM $\omega^2 = k/m$ × **Not on spec** SHM is not on the specification.

$Q = It$ $\quad charge = current \times time$ ✓ **On spec** On the specification equations list (3.3).

$P = I^2R$ & $W = I^2Rt$ ✓ **On spec** Both on the specification equations list (3.3).

# Modern physics

All modern physics questions are concerning changes in energy and therefore will be governed by one of the following four equations only

| | |
|---|---|
| $E = mc^2$ | Energy turning into mass or vice versa |
| $E = qV\ (W = QV)$ | Energy gained by a charged particle accelerated by a potential difference |
| $E = \frac{1}{2}mv^2$ | Kinetic energy of a particle |
| $E = hf$ | Energy of a photon |

E.g. photon producing a electron positron pair => $hf = m_e c^2 + m_{e^-} c^2 + \frac{1}{2}m_e v_e^2 + \frac{1}{2}m_{e^-} v_{e^-}^2$

Pair annihilation => $m_e c^2 + m_{e^-} c^2 = 2hf$

Particle accelerator => $qV = 1/2mv^2$ etc……..

Mass energy conserved, momentum conserved, charge conserved.

Radioactive decay:

Alpha emission

Momentum conserved means,

$$m_1 v_1 = m_2 v_2 \Rightarrow \frac{m_1}{m_2} = \frac{v_2}{v_1}$$

Ratio of velocities is the inverse of the ration of the masses

Ratio of K.E.

$$\frac{K.E_1}{K.E_2} = \frac{\frac{1}{2}m_1 v_1^2}{\frac{1}{2}m_2 v_2^2} = \frac{m_1 v_1 v_1}{m_2 v_2 v_2} = \frac{v_1}{v_2} \qquad (\ conservation\ of\ mometum \rightarrow m_1 v_1 = m_2 v_2\ )$$

Ratio of K.E is same as ratio of velocities which is inverse of ratio of the masses

# Derivations

$$\text{Equation 1: } v = u + at: \qquad \text{comes from the definition of acceleration: } a = \frac{\Delta v}{\Delta t} = \frac{v-u}{t}$$

$$\text{Equation 4: } s = \left(\frac{u+v}{2}\right)t \qquad \text{comes from definition of average speed}$$

$$\text{Equation 2: } s = ut + \frac{1}{2}at^2: \qquad \text{Use Equation 1 \& Equation 4 to get rid of } v$$

$$\text{Equation 3: } v^2 = u^2 + 2as: \qquad \text{Use Equation 1 \& Equation 4 to get rid of } t$$

---

Newton's 2nd Law: Force is proportional to rate of change of momentum (see definition of impulse is needed)

$$F \alpha \frac{\Delta p}{\Delta t}$$

$$F = k\frac{\Delta p}{\Delta t} = k\frac{mv - mu}{t} = km\frac{(\,v-u\,)}{t} = kma \qquad \text{by definition of the Newton } k = 1$$

**? Check** Deriving F = ma from momentum is not listed in the specification: check.

---

$$s = r\theta$$
$$\frac{s}{t} = \frac{r\theta}{t}$$
$$v = \omega r$$

**? Check** v = rω is not on the specification list: check.

---

$$W = mg = G\frac{Mm}{d^2}$$
$$g = G\frac{M}{d^2}$$

---

For satellite motion gravitational for between satellite and orbited body is the force due to gravity

$$\frac{mv^2}{r} = G\frac{Mm}{r^2}$$
$$v^2 = \frac{GM}{r}$$

---

Contents

# Derivations

Keplers 3rd Law

$$\frac{mv^2}{r} = G\frac{Mm}{r^2}$$
$$v^2 = \frac{GM}{r}$$
$$T = \frac{2\pi r}{v}$$
$$T^2 = \frac{4\pi^2 r^3}{GM}$$

---

Hooke's Law

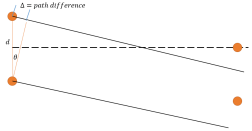
$$F = -ks$$

Get to

$$a = -\omega^2 s$$
$$\frac{F}{m} = -\frac{k}{m}s$$

× **Not on spec** SHM is not on the specification.

---




From right angled triangle, $\Delta = d \sin\theta$

For constructive interference to occur $\Delta = n\lambda$

$$n\lambda = d \sin\theta$$

**? Check** Grating equation not on the specification list: check.

---

# Derivations

For resistors in series, current is the same and voltage splits

$$V = V_1 + V_2$$
$$I = I_1 = I_2$$
$$Apply\ Ohm's\ Law: V = IR$$
$$IR = I_1 R_1 + I_2 R_2$$
$$R = R_1 + R_2$$

---

For resistors in series, voltage is the same and current splits

$$V = V_1 = V_2$$
$$I = I_1 + I_2$$
$$Apply\ Ohm's\ Law: V = IR$$
$$\frac{V}{R} = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

---

Get to

$$F = IlB$$

$$F = vqB$$
$$I = \frac{Q}{t}$$

---

# New specification check

Annotated October 2026 against the Leaving Certificate Physics specification introduced in September 2025 (first exam 2027). Only yellow-highlighted (course) material is checked, plus content new to the specification. Original notes are unchanged; coloured notes are added under each entry.

**Key:** ✓ On spec ✕ Not on spec ? Check + New

## Not on the new specification

- Impulse
- Moments and couples
- Simple harmonic motion
- Boyle's law
- Spherical mirrors (and the concave-mirror experiment), power of a lens, lenses in contact

## Check

- Radians, $v = r\omega$, centripetal acceleration; deriving $F = ma$ from momentum; magnification; diffraction grating; speed of sound experiment; frequency vs tension; photoelectric equation
- Wheatstone bridge, Joule's law experiment, $F = IlB$, $F = qvB$, $\Phi = BA$, capacitors; depth of particle physics

## New on the specification

- Vectors (components, resultants); 2D collisions; elastic PE $\frac{1}{2}kx^2$; escape velocity; four models for g
- Thermometric properties, heat pump, insulation; EM spectrum, dispersion, solar irradiance, ultrasound; $I = P/A$
- Fuses, p-n junction, n-p-n transistor, DC motor, generators, National Grid; thermionic emission, X-rays, emission spectra ($E_2 - E_1 = hf$), Cockcroft and Walton

## Sources and caveats

Strands 1–2: final specification (curriculumonline.ie) and its equations list. Strands 3–4: the final text could not be extracted, so these use the equations list plus the December 2023 NCCA draft. Confirm "?" items against the final specification. The specification has its own equations list; check whether a revised Formulae and Tables booklet changes the page numbers.

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